



WI-2-60-1-6

JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND TECHNOLOGY
University Examinations 2016/2017

**YEAR III SEMESTER I EXAMINATION FOR THE DEGREE OF BACHELOR OF
INFORMATION TECHNOLOGY**

BIT 2111: COMPUTER AIDED DESIGN

DATE: DECEMBER, 2016

TIME: 2 HOURS

INSTRUCTIONS: Answer Question ONE and Any Other TWO Questions.

Question One – 30 Marks

- a) Briefly discuss each of the following terms as applied in computer aided design:
- i. Hatch [1 mark]
 - ii. Layer [1 mark]
 - iii. Block [1 mark]
- b) Outline six advantages of using blocks. [6 marks]
- c) Discuss how you can close a file with CAD without leaving autoCAD. [3 marks]
- d) Explain how to add a new layer in autoCAD. [6 marks]
- e) Outline five dimensioning styles. [5 marks]
- f) Using a well labeled diagram, explain the elements of a dimension. [5 marks]
- g) What is the use of STARTUP and FILEDIA system variables. [2 marks]

Question Two

- a) Describe how to add text in CAD. [3 marks]
- b) Explain the following CAD terminology. [2 marks each]
- i. Grud

- ii. Cross hairs
- iii. Limits
- iv. Plat
- v. Snap

c) Using examples, describe three systems of entering points in autoCAD. [9 marks]

Question Three

a) Discuss the following tools. [2 marks each]

- i. CAE
- ii. CADD
- iii. CIM
- iv. CAM
- v. MRP
- vi. FEM
- vii. PDM

b) Explain three objectives MRP is meant to meet. [6 marks]

Question Four

a) Explain four types of projections in technology drawing. [8 marks]

b) Explain how you use a rectangle command in autoCAD. [5 marks]

c) Discuss how to display grid and set grid spacing. [4 marks]

d) How do you turn automatic save feature on or off in autoCAD. [3 marks]

Question Five

a) Outline eight advantages of using CAD design systems. [8 marks]

b) Outline how to turn on snap mode and set snap spacing. [6 marks]

c) Differentiate between save, save as and save in CAD. [2 marks]

d) Explain the two measurement system variable in autoCAD when you start a drawing from scratch. [4 marks]